Focused Mechanistic Hypothesis Search: MDD–Vitiligo p38α/MAPK14 Shared Mediator

DisMech issue #5921 — Focused hypothesis audit Hypothesis ID: `mdd_vitiligo_acquired_mapk14_shared_mediator` Disease scope: Major depressive disorder (MONDO:0002009) and vitiligo (MONDO:0008661, acquired generalized/nonsegmental prioritized) Literature coverage through 2026-07-26 (PubMed via provided tool). All snippets are **curator-verification leads**, not validated facts. No primary GEO re-analysis was performed (datasets not provided).

Executive Judgment

Overall verdict: REFUTED as a shared mediator / UNRESOLVED as a knowledge gap. Current evidence does **not** support an acquired, MAPK14-specific, cell-state-resolved causal mediator shared by MDD and vitiligo. The seed claim rests on a single, explicitly hypothesis-generating in-silico study, and every stronger human test points away from a shared p38α circuit. The observed comorbidity is better explained by **parallel tissue-specific p38 use plus nonspecific inflammatory and psychosocial burden**.

Stated separately for the three required sub-claims:

Sub-claim	Verdict	Basis
1. MAPK14 involvement in MDD	**Weakly supported / partly contradicted**	Only isoform-specific p38α evidence is a mouse serotonergic-neuron knockout (PMID:21835346); the human p38-inhibitor RCT (losmapimod) was **null** (PMID:24699061).
2. MAPK14 involvement in vitiligo	**Partially supported (pan-p38, not isoform-resolved)**	Human perilesional keratinocyte p38 activation (PMID:20085492), but pan-p38, not MAPK14-specific; MAPK14 is not a recognized vitiligo susceptibility gene (PMID:39890561).
3. A shared MAPK14-dependent mediator of comorbidity	**Refuted / unsupported**	No comorbid cohort, no paired tissue, no colocalization, null directional MR (PMID:37975615), isoform/species discordance, and nonspecific comorbidity matching atopic dermatitis (PMID:41781039).

Curation recommendation: Keep the pair-level hypothesis as a **KNOWLEDGE_GAP discussion**, optionally narrowed to **parallel tissue-specific p38 models**. The evidence does **not** justify a MAPK14 pathophysiology node, biomarker, causal edge, or conserved module in DisMech.

Summary

The seed hypothesis proposes that an acquired, cell-state-specific inflammatory program involving p38α/MAPK14 acts as a *shared mediator* linking major depressive disorder and vitiligo. A genuine shared mediator would require evidence that one defined MAPK14-dependent state — or a causally connected cross-compartment program — precedes and contributes to *both* disease outcomes. Over five iterations of literature audit and evidence grading, no such bridge was found.

The entire molecular basis for the shared-mediator claim traces to a single computational paper (PMID:42418414) that integrated **separate** disease datasets — MDD whole blood (GSE98793: 128 MDD/64 control) and vitiligo skin (GSE65127+GSE53146: 15/15) — using a permissive differential-expression threshold (any nonzero |log2FC|, nominal P<0.05), STRING PPI centrality, machine-learning feature selection, and ssGSEA-*inferred* (not measured) immune enrichment. In that paper MAPK14 is only **one of three co-equal machine-learning**

candidates (alongside EXOSC7 and KLRG1), the authors explicitly frame it as preliminary and hypothesis-generating, and there is no comorbid cohort, no paired blood/skin/brain material, no direct p38α activity assay, and no causal perturbation. This is a correlated-marker-in-separate-tissues result, not a mediation result.

Every independent line of evidence that could upgrade the bridge edge instead weakens it. The two disease arms use p38 in **isoform- and species-discordant** ways: the only isoform-specific p38α (MAPK14) finding is a mouse serotonergic-neuron knockout that produces stress *resilience* via serotonin-transporter trafficking (PMID:21835346), whereas the human vitiligo evidence is **pan-p38** apoptotic signaling in perilesional keratinocytes (PMID:20085492). A controlled human trial of the p38α/β inhibitor losmapimod was **negative** in MDD (PMID:24699061); bidirectional Mendelian randomization between generalized vitiligo and depression is **null** (PMID:37975615); MAPK14 is **not** among the >50 established vitiligo susceptibility loci (PMID:39890561); and vitiligo's mental-health comorbidity is **nonspecific**, closely matching atopic dermatitis (PMID:41781039, PMID:42051022). Meanwhile the vitiligo→MDD arm is well explained by **psychosocial mediation**, with depression burden scaling with lesion visibility, extent, and younger age (PMID:34554406, PMID:42082425), mirroring the age-stratified vitiligo→MDD risk gradient in the epidemiological cohort (PMID:30528503).

Taken together, the comorbidity is real and bidirectional at the epidemiological level, but the proposed molecular bridge is unsubstantiated. The best-supported models are **parallel tissue-specific p38 use** (p38α with different upstream triggers, cell types, substrates, and consequences in brain vs skin) layered on **nonspecific inflammatory/psychosocial burden**, not a single shared MAPK14-dependent mediator.

Key Findings

F001 — The seed molecular link rests on one hypothesis-generating in-silico study

The MAPK14 shared-mediator idea originates entirely from PMID:42418414 (2026), which integrated separate GEO datasets: MDD whole-blood (GSE98793: 128 MDD, 64 control) and vitiligo skin (GSE65127+GSE53146: 15 vitiligo, 15 control). MAPK14 was selected through a chain of correlational/heuristic steps — differential expression at a permissive threshold, STRING PPI network centrality, machine-learning feature selection, and ssGSEA immune-enrichment signatures that are *inferred* rather than measured cell abundances. Critically, MAPK14 is one of **three** co-equal machine-learning hits: the paper states, *"Machine learning further prioritized three key genes: EXOSC7, KLRG1, and MAPK14."* The authors themselves characterize the result cautiously: *"Preliminary validation suggests MAPK14 as a potential candidate gene warranting further investigation in MDD and vitiligo."*

There is no comorbid MDD–vitiligo cohort, no paired samples from the same participants, no direct p38α activity assay, and no causal perturbation. The comparison also confounds tissue (whole blood vs skin), so any "shared" signal may simply track leukocyte composition. MAPK14 is therefore not uniquely implicated even within its source paper, and none of the inference steps (coexpression, PPI centrality, ML selection, ROC performance, ssGSEA, pathway enrichment) can establish causality or mediation. EXOSC7 and KLRG1 were co-equal hits and were not independently audited — they should be evaluated on equal footing before MAPK14 is privileged.

F002 — Bidirectional epidemiological association is real, but antidepressant use *lowered* vitiligo risk

The population-based cohort (PMID:30528503, THIN database) establishes temporal association in both directions: MDD→vitiligo adjusted HR **1.64** (95% CI 1.43–1.87, P<.0001; n=405,397 MDD vs 5,739,048 comparators), and vitiligo→MDD HR **1.31** (before age 30) and **1.22** (age ≥30). The abstract states MDD patients *"were at a 64% increased risk for vitiligo (hazard ratio 1.64, 95% confidence interval [CI] 1.43-1.87, P < .0001)."*

Crucially, the same study reports that vitiligo risk *"was decreased in patients using antidepressants."* Antidepressants (which in vitro *modulate* p38, PMID:40840361) reducing vitiligo risk is more consistent with psychosocial/treatment models than with a shared pro-p38 inflammatory program that antidepressants would be expected to leave unchanged or worsen. This is association and temporal ordering of recorded diagnoses — **not mediation**.

F003 — Human vitiligo p38 and MDD p38α evidence are isoform- and species-discordant

The two arms do not use the same molecular entity in the same way:

- **Vitiligo arm (human, pan-p38):** PMID:20085492 shows activated p38 as an upstream driver of apoptosis in perilesional keratinocytes from 12 human nonsegmental-vitiligo patients — *"our study demonstrates the pivotal role of p38 MAP kinase as an upstream signal of perilesional keratinocyte damage"* — but this is **pan-p38**, not MAPK14-isoform-resolved.

Supporting melanocyte models (PMID:17481858, PMID:21514118) show dopamine/oxidative-stress-induced JNK/p38 apoptosis, again pan-p38 and partly mouse (Mel-Ab) cells.

- **MDD arm (mouse, p38α-specific):** PMID:21835346 is the *only* MAPK14/p38α-isoform-specific result — *"the α isoform of p38 mitogen-activated protein kinase (MAPK) was selectively inactivated"* — but selective p38α deletion in **mouse serotonergic neurons** produces stress *resilience* via SERT trafficking/hyposerotonergic state. This is rodent depression-like behavior, not human MDD, and it is in neurons, not blood or skin.

Directional genetics is also null: PMID:37975615 found *"none of the rigorous bidirectional MR analyses uncovered a significant causal association"* between generalized vitiligo and depression. The arms therefore share a *gene family label* but differ in isoform specificity, species, cell type, upstream trigger, downstream substrate, and directional consequence — the signature of **parallel tissue-specific p38 use**, not one shared circuit.

F004 — A p38 inhibitor (losmapimod) failed in a controlled human MDD trial

The strongest causal test in the MDD arm is negative. PMID:24699061 evaluated losmapimod (GW856553, an oral p38α/β inhibitor) 7.5 mg BID for 6 weeks in two RCTs in clinically diagnosed MDD enriched for anergia/psychomotor retardation. An underpowered, early-terminated Study 574 (n=24) nominally favored drug (Bech difference −4.10; 95% CI −7.36, −0.83; p=0.017), but the pre-registered Bayesian confirmatory Study 009 (n=128) *"showed no advantage for losmapimod"* (Bech difference +1.11; 95% credible interval −0.22, 2.50), with no significant cytokine biomarker changes. The authors conclude: *"In conclusion 7.5 mg BID losmapimod was not effective in MDD."* Because losmapimod inhibits p38α (MAPK14) — albeit not isoform-selectively — this is direct human evidence against a *causal, druggable* p38 role in MDD.

F005 — Vitiligo's mental-health comorbidity is largely nonspecific, matching atopic dermatitis

If a vitiligo-specific p38 mediator drove MDD, vitiligo's psychiatric burden should exceed that of other inflammatory dermatoses. It does not. PMID:41781039 (German DAK claims, N=2,885,984; 4,631 vitiligo cases, ICD-10 L80, propensity-matched 1:3) reports depressive-episode prevalence of 8.9–19.2% and, in matched comparison with atopic dermatitis, *"only a few and inconsistent differences were observed"* (more pronounced differences appeared only vs psoriasis). The companion claims analysis PMID:42051022 reports modest, shared risk: depression RR **1.23** (95% CI 1.15–1.32), anxiety RR **1.32** (1.19–1.47), explicitly noting these were *"linked to skin diseases like vitiligo, atopic dermatitis, or psoriasis."* Modest effect sizes (RR ~1.2–

1.3) shared across dermatoses support **nonspecific inflammatory/psychosocial burden** rather than a vitiligo-specific molecular mechanism. Note these two claims analyses share the DAK data source and likely overlapping cohorts — treat as **one data source**, not two independent replications.

F006 — No genetic evidence for MAPK14 as a shared pleiotropic locus

Vitiligo susceptibility is immune/oxidative/melanogenic, not MAPK14-centered. The genetics review PMID:39890561 states GWAS *"have identified over 50 susceptibility loci, including key genes within the MHC region and those involved in immunity, oxidative stress, and melanogenesis"* — MAPK14 is not among the highlighted genes. The newest multi-omics vitiligo mediator is TAPBP via STAT2/interferon–antigen-presentation (PMID:41884389): *"Cross-omic analysis identified TAPBP as a top candidate, which is significantly upregulated in lesional melanocytes and core to the antigen-processing network"* — again, not MAPK14. Targeted PubMed searches (2026-07-26) for an MDD–vitiligo cross-trait genetic-correlation/colocalization/LDSC study and for a MAPK14 depression GWAS locus returned no papers, and the only genetic disease-to-disease test (bidirectional MR, PMID:37975615) was null. There is currently no locus-level colocalization, shared eQTL/pQTL, or genetic-correlation evidence placing MAPK14 as a shared pleiotropic driver. (Null MR does not, by itself, exclude shared pleiotropy — but no cross-trait analysis supports it either.)

F007 — Psychosocial mediation is well-supported for the vitiligo→MDD arm

The systematic review PMID:34554406 (168 studies, 1979–2021) reports depression in 0.1–62.3% and anxiety in 1.9–67.9% of vitiligo patients, with significantly higher burden associated with *"female sex, visible or genital lesions, age < 30 years (particularly adolescents), and greater body surface area involvement, among others,"* and lesion concealment as the commonest coping strategy. The narrative review PMID:42082425 (2026) concurs that *"Disease visibility, particularly involvement of the face or genital regions; greater body surface area involvement; active disease progression; and longer disease duration emerged as key clinical factors associated with worse outcomes."* This visibility/age gradient mirrors the age-stratified vitiligo→MDD risk (HR 1.31 if <30y vs 1.22 if ≥30y) in PMID:30528503, consistent with **psychosocial rather than molecular mediation**.

Mechanistic Model / Interpretation

The evidence supports **two independent (parallel) p38 pathways** plus a **psychosocial/nonspecific-inflammation overlay**, not a single shared mediator.

```
BRAIN / NEURAL COMPARTMENT (MDD arm)
  chronic stress / neuroinflammation
       │
       ▼
  serotonergic neurons ──[p38α / MAPK14, isoform-specific]──► SERT trafficking → hyposerotonergi
       │  (MOUSE ONLY, deletion → RESILIENCE; <a href="https://pubmed.ncbi.nlm.nih.gov/21835346/"
       ▼
  depression-like behavior  ── human MDD?  ◄─ p38 inhibitor losmapimod NULL in RCT (<a href="htt

SKIN COMPARTMENT (vitiligo arm)
  oxidative stress (H2O2 / dopamine)
       │
       ▼
  melanocytes + perilesional keratinocytes ──[pan-p38, NOT isoform-resolved]──► p53/NF-κB apopto
       │  (HUMAN NSV keratinocytes + mouse/human melanocytes; <a href="https://pubmed.ncbi.nlm.ni
       ▼
  melanocyte loss → depigmentation

BRIDGE EDGE (proposed shared mediator)
  ??? shared upstream state / circulating mediator / homologous cell program ???
       │
       ▼
  ✗ MISSING: no comorbid cohort, no paired tissue, no colocalization,
    null directional MR, isoform/species discordance

PSYCHOSOCIAL / NONSPECIFIC OVERLAY (best-supported real link)
  visible/extensive vitiligo, younger age, stigma → distress → MDD
  (<a href="https://pubmed.ncbi.nlm.nih.gov/34554406/" rel="noopener noreferrer" title="Visit Pu
```

Causal-chain grading:

Edge	Grade	Note
MDD: stress → serotonergic neuron → p38α → SERT → behavior	**Partial (mouse)**	Isoform-specific but rodent; deletion→resilience complicates a "p38α drives MDD" story
MDD: p38 activity → clinical MDD outcome	**Contradicted**	Losmapimod RCT null (P 24699061)
Vitiligo: oxidative stress → keratinocyte/melanocyte → p38 → apoptosis	**Supported (pan-p38)**	Human NSV, but not MAPK14-isoform-resolved
Vitiligo: MAPK14-specific necessity	**Missing**	No isoform-selective perturbation/rescue
Bridge: shared MAPK14-dependent state	**Missing / Speculative**	No direct human bridge evidence
Shared pleiotropy at MAPK14	**Missing**	Not a vitiligo locus; no cross-trait colocalization
Psychosocial mediation (vitiligo→MDD)	**Supported**	Visibility/age/extent gradient

Cell-state map — is any state truly shared? The neural substrate (SERT trafficking in serotonergic neurons) and the skin substrate (p53/NF-κB apoptosis in melanocytes/keratinocytes) are **analogous** (both engage p38 family kinases under stress) but **not shared**: different upstream triggers (psychosocial stress vs oxidative/dopamine), different cells, different substrates, and different isoform resolution. The circulating-immune "bridge" candidate (whole-blood MAPK14 transcript, P 42418414) is speculative and confounded by leukocyte composition. **No truly shared MAPK14-dependent state is demonstrated.**

Evidence Base

PMID	Design & type	Species / phenotype	Size	Tissue / cell	MAPK14 measure/ perturbation	Result & direction
42418414	In-silico integration (DE/PPI/ML/ ssGSEA)	Human MDD (blood) + vitiligo (skin), separate	128/64 + 15/15	Whole blood vs skin	Transcript DEG + ML selection	MAPK14 = 1 of 3 co-equal ML hits; nominal P
30528503	Bidirectional cohort	Human MDD & vitiligo (dx)	405,397 MDD; 5.7M comp.	Registry	None	MDD→vitiligo HR 1.64; vitiligo→MDD 1.31/1.22; antidepressants ↓ risk
24699061	Two RCTs (Bayesian)	Human clinical MDD	24 + 128	Systemic	p38α/β inhibitor losmapimod	Confirmatory NO advantage; "not effective in MDD"
21835346	Conditional KO	Mouse depression-like	—	Serotonergic neurons	Selective p38α (MAPK14) deletion	Deletion → resilience via SERT
20085492	Ex vivo mechanistic	Human NSV	12	Perilesional keratinocytes	Pan-p38	p38 upstream of apoptosis
17481858	In vitro	Mouse Mel-Ab + human melanocytes	—	Melanocytes	Pan JNK/p38	Dopamine → apoptosis
21514118	In vitro	Human melanocytes	—	Melanocytes	Pan p38/JNK/ Akt	Dopamine → apoptosis; apigenin protective
37975615	Bidirectional MR	Generalized vitiligo & depression	GWAS-scale	Genetic	Instruments (disease-level)	No significant causal association

PMID	Design & type	Species / phenotype	Size	Tissue / cell	MAPK14 measure/ perturbation	Result & direction
41781039	Matched cohort (claims)	Vitiligo (L80) vs AD/ psoriasis	4,631 cases	Registry	None	Profile ≈ atopic dermatitis
42051022	Claims analysis	Vitiligo	Population	Registry	None	Depression RR 1.23; anxiety RR 1.32
39890561	Review (genetics)	Vitiligo	—	Genetic	None	>50 loci: MHC/ immune/ oxidative/ melanogenic; not MAPK14
41884389	Multi-omics + experiment	Vitiligo	—	Lesional melanocytes	None	TAPBP/STAT2/ IFN top mediator
34554406	Systematic review	Vitiligo psychosocial	168 studies	—	None	Burden ↑ with visibility, extent, age<30
42082425	Narrative review	Vitiligo QOL	—	—	None	Visibility/ extent/activity drive QOL
40840361	In vitro	Macrophages	—	Macrophages	Phospho-p38 (flow)	Antidepressants modulate p38 (context-dependent)

Rodent neuroinflammation p38 papers (PMID:41707798, PMID:41786069, PMID:42134655) reinforce a neural p38 route in mouse depression-like behavior but are pan-p38/pan-MAPK and cannot be upgraded to human MDD or a shared mediator.

P 42418414

Replication / Method Audit

No independent GEO re-analysis was performed (datasets not supplied). This is a methodological audit of the seed design as reported.

Element	GSE98793 (MDD)	GSE65127 / GSE53146 (vitiligo)	Assessment
Tissue	Whole blood	Skin	**Cross-tissue mismatch** — cannot establish a shared cell-state
Disease definition	MDD (clinical)	NSV / vitiligo as annotated	Subtype/activity/treatment not harmonized
N	128 / 64	15 / 15 (integrated)	Vitiligo side very small; integration adds batch risk
DE threshold	nonzero |log2FC|, nominal P<0.05	same	**No effect-size floor, no FDR reported**
Multiple-testing control	Not evident	Not evident	Unsupported without FDR
Feature-selection independence	ML on same DEGs	—	**Validation likely not independent** → optimistic
Immune cells	ssGSEA (inferred)	ssGSEA (inferred)	Not measured; tracks leukocyte composition
MAPK14 direction/ CI/adj-P	Not reported in abstract	—	**Unverified** — curators must extract from full text
Comorbid/paired samples	None	None	**No bridge between diseases**

Reproduces independently: MDD–vitiligo comorbidity exists (P 30528503); p38 is stress-activated in immune cells and skin (non-specific). **Does NOT reproduce:** MAPK14 as a *specific* shared driver over EXOSC7/KLRG1 or generic inflammatory markers; any shared cell-state; MAPK14 direction/magnitude/adjusted significance; any causal or mediation claim. GSE52790 and GSE80009 were not examined (not provided) — flagged as open replication tasks.

Model-Comparison Table

Model	Causal prediction	Best support	Best contradiction	Distinguishing finding	Plausibility
Shared MAPK14 mediator	One MAPK14 state precedes both	P 42418414	Null RCT (24699061), null MR (37975615), isoform/species discordance	Paired blood+skin phospho-p38α shared state in comorbid patients	**Low**
Psychosocial mediation	Visibility/ stigma → MDD	34554406, 42082425, 30528503 age gradient	Some less-visible cases still affected	Link persists/ attenuates after severity/ distress adjustment	**High (vitiligo→MDD)**
Surveillance / treatment	Diagnostic contact / drugs inflate	Antidepressants ↓ vitiligo risk (30528503)	Bidirectionality survives adjustment	Negative controls, equalized follow-up	**Moderate**
Nonspecific inflammation	Comorbidity shared across dermatoses	41781039 (≈AD), 42051022	Some signals stronger vs psoriasis	Signature distinguishing vitiligo-MDD from AD-MDD	**High**
Shared pleiotropy	Common liability at shared loci	—	Not a vitiligo locus (39890561); null MR (37975615)	Cross-trait LDSC + locus colocalization	**Low (for MAPK14)**
MDD→vitiligo causation	MDD/stress worsens vitiligo	HR 1.64 (30528503)	Null MR (37975615); antidepressants protective	Strong-instrument MR; prospective incidence	**Weak–moderate**
Vitiligo→MDD causation	Vitiligo causes MDD	HR 1.31/1.22 (30528503)	Null MR (37975615); nonspecific vs AD	Mediation adjusting for visibility/ stigma	**Moderate (psychosocial route)**

Model	Causal prediction	Best support	Best contradiction	Distinguishing finding	Plausibility
Parallel tissue-specific p38	p38α used separately in brain vs skin	Discordance (21835346 vs 20085492)	— (consistent with model)	Refuted only by a demonstrated shared cross-compartment program	**High (best fit)**

Limitations and Knowledge Gaps

1. **No comorbid human cohort exists** with clinically adjudicated MDD *and* subtype/activity-defined vitiligo carrying MAPK14 RNA, p38α protein, phospho-p38 activity, or downstream substrate measurements. This is the central missing bridge study; its absence is documented from targeted PubMed searches, not a comprehensive multi-database systematic review.
2. **Isoform specificity is unresolved** in vitiligo: all skin evidence is pan-p38, so MAPK14 (p38α) vs MAPK11/12/13 contributions are unknown. No isoform-selective genetic perturbation or kinase-dead rescue exists for either arm's human context.
3. **Transcript ≠ protein ≠ activity:** the seed relies on bulk transcript abundance; p38α protein, phosphorylation, and substrate activity were never measured across compartments.
4. **The MDD isoform-specific evidence is mouse-only** and points to resilience-on-deletion, complicating a straightforward "p38α drives MDD" narrative.
5. **Confounders** — cell mixture, medications, disease activity, batch, healthcare contact, smoking, BMI, socioeconomic factors — are largely uncontrolled in the seed and unaddressed for the molecular hypothesis.
6. **Null MR does not exclude shared pleiotropy** — no cross-trait LDSC/colocalization has been performed, so the pleiotropy model is untested rather than refuted (though MAPK14-specific pleiotropy is unsupported by known vitiligo genetics).
7. **Search scope:** single database (PubMed); Embase, GWAS Catalog, Open Targets, ClinicalTrials.gov, and preprint servers were not queried. "No evidence found" applies only to explicitly searched scopes.

Proposed Follow-up Experiments / Discriminating Studies

Ranked by efficiency in separating the competing models:

1. **Cross-trait LDSC + MAPK14-locus colocalization (highest value, lowest cost).** Vitiligo GWAS × MDD GWAS global/local genetic correlation, locus colocalization, and shared eQTL/pQTL at MAPK14, distinguishing colocalization from linkage and sample overlap. Summary statistics already exist. *Expected under shared pleiotropy:* colocalization at MAPK14; *under parallel/nonspecific:* none.
2. **Comorbid paired-tissue profiling.** Treatment-naive MDD-only, active-NSV-only, comorbid, matched disease-free, and a visible-inflammatory-skin comparator (atopic dermatitis). Single-cell + surface-protein/**phospho-p38α** (CyTOF) in blood and paired lesional/nonlesional skin. *Under shared mediator:* one phospho-p38α+ state elevated in both arms, maximal in comorbid subjects; *under parallel/nonspecific:* compartment-specific signatures indistinguishable from atopic dermatitis.
3. **Isoform-resolved perturbation with rescue.** Cell-restricted MAPK14 CRISPR knockout/interference in human iPSC-derived serotonergic neurons/glia and in melanocytes/keratinocytes/CD8 T cells, with selective p38α pharmacology plus CRISPR-resistant **wild-type vs kinase-dead** MAPK14 rescue and dose-response. Establishes isoform specificity, causal necessity, and on-target vs off-target effects per compartment.
4. **Longitudinal mediation with equalized surveillance.** Prospective repeated phenotyping measuring MAPK14/phospho-p38 and cytokines *before* onset of the second condition, with mediation adjusting for visibility/severity, stigma, stress, socioeconomic factors, smoking, BMI, autoimmune comorbidity, medications, and healthcare utilization; include negative-control outcomes and active comparators to address surveillance/treatment bias.
5. **Vitiligo activity RCT of a selective p38α inhibitor** with a phospho-p38 target-engagement biomarker, mirroring the informative null MDD RCT to test the vitiligo-arm causal edge.

Prespecified decision criteria: Shared mediation requires a demonstrated shared MAPK14+ cell-state (study 2) **and** MAPK14-locus colocalization (study 1) **and** isoform-specific necessity in both arms (study 3). Absence of a shared state but arm-specific necessity → **parallel p38**. Vitiligo≈AD signature → **nonspecific**. Effect abolished by visibility/stigma adjustment → **psychosocial**. Attenuation by equalized surveillance → **surveillance bias**. Null coloc + null MR → reject shared-pleiotropy/directional-genetic.

Curation Leads (require independent DisMech verification)

- **Disposition:** Keep `mdd_vitiligo_acquired_mapk14_shared_mediator` as a **KNOWLEDGE_GAP discussion**, optionally **narrowed to parallel tissue-specific p38 models.** Do **not** promote to EMERGING mechanistic hypothesis; consider deprecating the "single shared mediator" framing.
- **Node/biomarker/edge/module:** Evidence does **NOT** justify a MAPK14 pathophysiology node, biomarker, causal edge, or conserved cross-disease module for this pair. At most, MAPK14 is a hypothesis-level candidate marker in one in-silico study.
- **Seed lead:** PMID:42418414 — *"Machine learning further prioritized three key genes: EXOSC7, KLRG1, and MAPK14."*
- **MDD refutation lead:** PMID:24699061 — *"In conclusion 7.5 mg BID losmapimod was not effective in MDD."*
- **Vitiligo refinement:** p38 evidence is pan-p38, ex-vivo/in-vitro (PMID:20085492); genetics point to HLA/immune/oxidative/TAPBP (PMID:39890561, PMID:41884389), not MAPK14.
- **Alternative candidates:** EXOSC7 and KLRG1 were co-equal ML hits and were not independently audited — evaluate on equal footing before privileging MAPK14.
- **Overlap flag:** PMID:41781039 and PMID:42051022 share the DAK claims source — treat as one data source.

Supported vs Refuted Hypotheses (summary)

- **Supported:** MDD–vitiligo comorbidity is real, bidirectional, and temporally ordered (P 30528503). p38 is genuinely activated in vitiligo skin (pan-p38, P 20085492) and p38α is mechanistically active in mouse serotonergic stress circuitry (P 21835346). Psychosocial mediation of vitiligo→MDD is well-supported

(
P 34554406
).
P 42082425

- **Refuted / contradicted (as stated):** (i) p38 inhibition treats human MDD — **refuted** (
P 24699061
).
(ii) Vitiligo has a *specific* MDD-linking signature beyond other skin diseases — **refuted** (
P 41781039
).
(iii) Genetic directional causation vitiligo↔depression — **null** (
P 37975615
).
(iv) MAPK14 as an established shared pleiotropic locus — **unsupported** (
P 39890561
).
P 41884389
- **Unresolved:** whether a narrow, cell-state-restricted MAPK14 program contributes to each disease **independently** (parallel model) remains open and testable.

Citation Manifest

Identifier	Title (abbrev.)	Year	Study type	Sections used
P 42418414	Immune microenvironment in MDD and vitiligo	2026	Computational (in-silico)	Exec, Summary, F001, Evidence, Audit, Leads
P 30528503	Vitiligo and MDD: bidirectional cohort	2019	Population cohort	Exec, Summary, F002, F007, Evidence, Models
P 24699061	Losmapimod in MDD (two RCTs)	2014	RCT	Exec, F004, Evidence, Models, Leads
P 21835346	Selective p38α deletion in serotonergic neurons	2011	Mouse conditional KO	Exec, F003, Evidence, Map
P 20085492	p38/apoptosis in perilesional vitiligo keratinocytes	2010	Human ex vivo	Exec, F003, Evidence, Map, Leads
P 17481858	Dopamine-induced melanocyte apoptosis (JNK/p38)	2007	In vitro	F003, Evidence
P 21514118	Apigenin/DA melanocyte p38/JNK/Akt	2011	In vitro	F003, Evidence
P 37975615	Vitiligo–mental disorders MR	2024	Mendelian randomization	Exec, F003, F006, Evidence, Models
P 41781039	Mental-health risk in vitiligo vs AD/psoriasis	2026	Matched cohort (claims)	Exec, F005, Evidence, Models, Leads
P 42051022	Vitiligo epidemiology/comorbidity (Germany)	2026	Claims analysis	F005, Evidence, Leads
P 39890561	Genetics and epigenetics in vitiligo	2025	Review	Exec, F006, Evidence, Leads
P 41884389	TAPBP mediator in vitiligo	2026	Multi-omics + experiment	F006, Evidence, Leads

Identifier	Title (abbrev.)	Year	Study type	Sections used
P 34554406	Psychosocial effects of vitiligo (systematic review)	2021	Systematic review	F007, Evidence, Models
P 42082425	QOL impairment in vitiligo	2026	Narrative review	F007, Evidence, Models
P 40840361	Duloxetine/venlafaxine on p38 MAPK	2025	In vitro	F002, Evidence
P 41707798	COG133 / p38 neuroinflammation	2025	Mouse	Evidence (context)
P 41786069	CUMS Th1/microglia/p38	2025	Mouse	Evidence (context)
P 42134655	let-7a-5p/MAP3K1/MAPK in depression model	2025	Mouse/in vitro	Evidence (context)

All identifiers, quotes, and statistics are drawn from the abstracts/snippets recorded during the investigation and remain leads for independent DisMech curator validation.